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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore-607001
B. Sc DEGREE EXAMINATION-APRIL 2021
BC612S-BIO TECHNOLOGY AND GENETIC ENGINEERING

Time: 3 hrs **Max Marks: 75**

SECTION-A (25X1=25)
Answer ALL the questions:

I. Choose the best answer (10X1=10)

1. ----------possess genes for their own transfer from one cell to another.
   a) F-plasmids b) R-plasmids c) Bacteriophages d)Stringent plasmids
2. A liposome mediated gene transfer is referred as --------
   a) Transduction b) Lipofection
   c) Microinjection d) Particle bombardment
3. The culture of native tissue that retains most of the *in vivo* histological features is ----------.
   a) Organ culture b) Cell culture c) both a and b d)none
4. Cells that retain their proliferative capacity throughout life are called as ----
   a)Finite cells b) Continuous cells c) Stem cells d) Tissue
5. ---------- was the first synthetic vaccine given for public use.
   a)Hepatitis B vaccine b) BCG vaccine
   c)HSV vaccine d) FMD vaccine
6. ----------is the animal of choice for most transgenic experiments.
   a)Cow b) Rabbit c)Sheep d)Mouse
7. A -------- is an excised piece of differentiated tissue.
   a)Explant b)Callus c)Transgene d)Suspension
8. ---------- is regarded as the father of plant tissue culture.
   a)Gautheret b)White c)Haberlandt d)Nobecourt

9. -------- are naked plant cells without cell wall but possess plasma membrane and other cellular components.
   a) Protoplasts b) Chloroplasts c) Diploids d) Haploids
10. --------- refers to the containment system for cultivation of prokaryotic cells.
   a) Fermenter b) Bioreactor
   c) Airlift bioreactor d) Packed bed bioreactor

II. Say true or false (5X1=5)

11. Cosmids are the vectors possessing the characteristics of both plasmid and bacteriophages..
12. Subunit vaccines are genetically modified pathogenic organisms that are made non-pathogenic.
13. Glutamine supplies carbon for the operation of citric acid cycle.
14. Synchronization of cultured cells is the unorganized existence of majority of cells in the same cell cycle phase simultaneously
15. Carbohydrates constitute the most predominant source of energy in fermentation.

III. Answer in one sentence (10X1=10)

11. Totipotency.
12. Organogenesis
13. Fermentation
14. Subunit vaccines
15. Shuttle vector
16. Ligases
17. DNA vaccines
18. Somatic hybridization

19. Haploid plants
20. Cell lines

SECTION-B (4X5=20)

Answer any FOUR of the following:

26. Write a short note on the media used for fermentation.
27. What are the enzymes involved in rDNA technology?
28. Explain somatic embryogenesis.
29. Give an account on culture media.
30. Transgenic mice – give a brief account.
31. Write a note on
    i) phage M13
    ii) cosmids

SECTION-C (3X10=30)

Answer Any THREE of the following:

32. Describe in detail the gene transfer methods:
    a) Microinjection
    b) Electroporation
33. Explain the technique devised for the primary culture of isolated tissues.
34. Write a note on the isolation, identification and uses of stem cells.
35. Describe the process of protoplast isolation,fusion and regeneration.
36. Give an account on the classification of fermentation process.

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